



**Laboratorium
Multimedia dan Internet of Things
Departemen Teknik Komputer
*Institut Teknologi Sepuluh Nopember***

Laporan Akhir Praktikum Jaringan Komputer

Konfigurasi Vlan, OSPF Multivendor

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1 Langkah-Langkah Percobaan

1.1 Praktikum 1: Konfigurasi Cisco Switch

Tujuan pada praktikum ini adalah mengonfigurasi Cisco Switch untuk membuat VLAN 10, 20, 30, dan 40, mengatur port menuju PC sebagai access port, serta mengatur port menuju Cisco Router sebagai trunk port.

1. Masuk ke terminal Cisco Switch.

```
Switch> enable
Switch# configure terminal
Switch(config)#
```

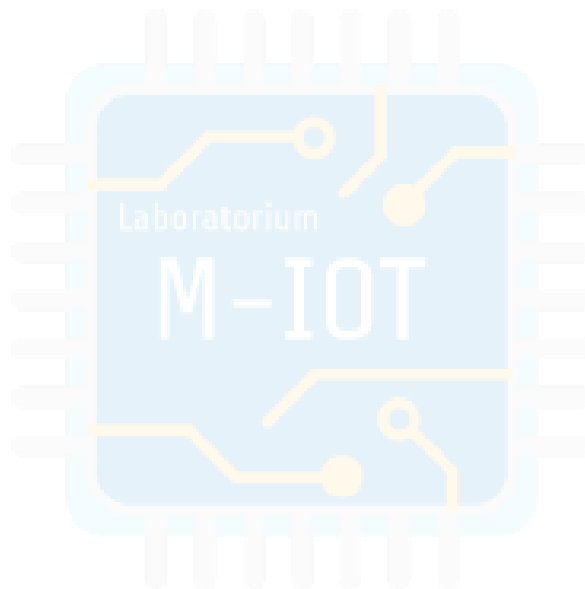
2. Membuat VLAN.

```
vlan 10
  name Finance
exit

vlan 20
  name HR
exit

vlan 30
  name IT
exit

vlan 40
  name Marketing
exit
```



3. Pastikan VLAN sudah dibuat dan statusnya aktif dengan perintah verifikasi berikut.

```
show vlan brief
```

```
Terminal
vIOS x Switch x

*****
* IOSv is strictly limited to use for evaluation, demonstration and IOS *
* education. IOSv is provided as-is and is not supported by Cisco's *
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Switch>show vlan brief

VLAN Name                Status    Ports
-----
1    default                active    Gi1/0, Gi1/2, Gi1/3
10   finance                 active    Gi0/3
20   HR                      active    Gi1/1
30   IT                      active    Gi0/1
40   marketing               active    Gi0/2
1002 fddi-default            act/unsup
1003 token-ring-default    act/unsup
1004 fddinet-default        act/unsup
1005 trnet-default          act/unsup
Switch>
```

Gambar 1: Verifikasi VLAN dengan perintah show vlan brief

4. Konfigurasi access port untuk setiap VLAN. Port access harus disesuaikan dengan interface pada Cisco Switch.

```
Switch#conf t

interface gi0/2
description PC-FINANCE-VLAN10
switchport mode access
switchport access vlan 10
no shutdown
exit

interface gi0/1
description PC-HR-VLAN20
switchport mode access
switchport access vlan 20
no shutdown
exit

interface gi1/1
description PC-IT-VLAN30
switchport mode access
switchport access vlan 30
no shutdown
```

```
exit
```

```
interface gi0/3
description PC-MARKETING-VLAN40
switchport mode access
switchport access vlan 40
no shutdown
exit
```

5. Konfigurasi trunk ke Cisco Router. Konfigurasi trunk port dari switch menuju Cisco Router untuk VLAN 10, 20, 30, dan 40 melalui interface Gi0/0.

Tabel 1: Konfigurasi Trunk Switch ke Cisco Router

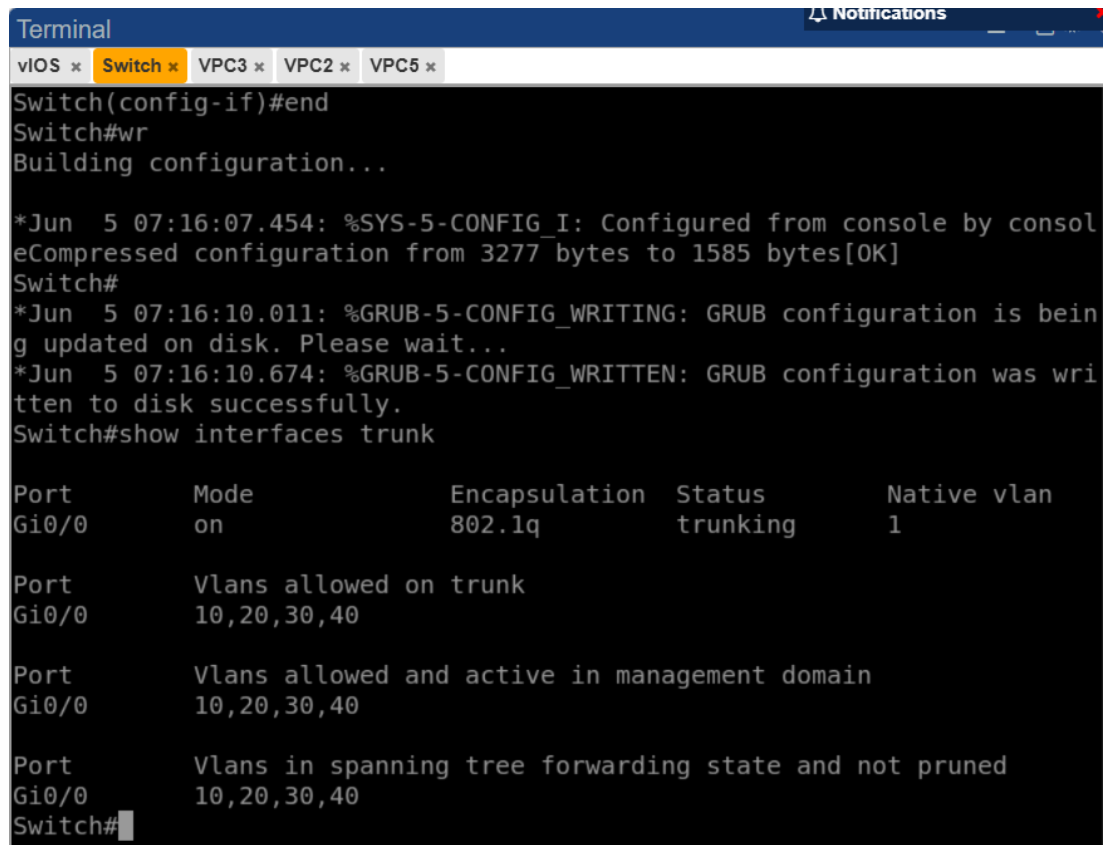
Interface Switch	Terhubung ke	Mode	VLAN yang Dibawa
Gi0/0	Cisco Router Gi0/2	Trunk	10,20,30,40

```
Switch#conf t
```

```
interface gi0/0
switchport trunk encapsulation dot1q
description TRUNK-TO-CISCO-ROUTER
switchport mode trunk
switchport trunk allowed vlan 10,20,30,40
no shutdown
exit
```

Verifikasi dan simpan konfigurasi:

```
show interfaces trunk
write memory
```



```
Terminal
vIOS x Switch x VPC3 x VPC2 x VPC5 x
Switch(config-if)#end
Switch#wr
Building configuration...

*Jun  5 07:16:07.454: %SYS-5-CONFIG_I: Configured from console by console
Compressed configuration from 3277 bytes to 1585 bytes[OK]
Switch#
*Jun  5 07:16:10.011: %GRUB-5-CONFIG_WRITING: GRUB configuration is being
updated on disk. Please wait...
*Jun  5 07:16:10.674: %GRUB-5-CONFIG_WRITTEN: GRUB configuration was written
to disk successfully.
Switch#show interfaces trunk

Port      Mode           Encapsulation  Status        Native vlan
Gi0/0     on             802.1q         trunking      1

Port      Vlans allowed on trunk
Gi0/0     10,20,30,40

Port      Vlans allowed and active in management domain
Gi0/0     10,20,30,40

Port      Vlans in spanning tree forwarding state and not pruned
Gi0/0     10,20,30,40
Switch#
```

Gambar 2: Verifikasi trunk dengan perintah show interfaces trunk

1.2 Praktikum 2: Konfigurasi Cisco Router

Tujuan pada praktikum ini adalah mengonfigurasi Cisco Router sebagai gateway untuk semua VLAN, mengonfigurasi DHCP server, serta mengimplementasikan ROAS (Router-on-a-Stick).

1. Masuk ke Cisco Router.

```
enable
configure terminal
```

2. Konfigurasi interface trunk ke Switch. Interface yang terhubung ke Cisco Switch adalah Gig0/2.

```
interface gig0/2
description TRUNK-TO-CISCO-SWITCH
no ip address
no shutdown
exit
```

3. Konfigurasi subinterface VLAN 10.

```
interface gig0/2.10
description GATEWAY-VLAN10-FINANCE
encapsulation dot1Q 10
ip address 192.168.10.1 255.255.255.0
no shutdown
exit
```

4. Konfigurasi subinterface VLAN 20.

```
interface gig0/2.20
description GATEWAY-VLAN20-HR
encapsulation dot1Q 20
ip address 192.168.20.1 255.255.255.0
no shutdown
exit
```

5. Konfigurasi subinterface VLAN 30.

```
interface gig0/2.30
description GATEWAY-VLAN30-IT
encapsulation dot1Q 30
ip address 192.168.30.1 255.255.255.0
no shutdown
exit
```

6. Konfigurasi subinterface VLAN 40.

```
interface gig0/2.40
description GATEWAY-VLAN40-MARKETING
encapsulation dot1Q 40
ip address 192.168.40.1 255.255.255.0
no shutdown
exit
```

7. Konfigurasi DHCP Server untuk VLAN 10.

```
ip dhcp excluded-address 192.168.10.1 192.168.10.10

ip dhcp pool VLAN10-FINANCE
network 192.168.10.0 255.255.255.0
default-router 192.168.10.1
dns-server 8.8.8.8
exit
```

8. Konfigurasi DHCP Server untuk VLAN 20.

```
ip dhcp excluded-address 192.168.20.1 192.168.20.10

ip dhcp pool VLAN20-HR
network 192.168.20.0 255.255.255.0
default-router 192.168.20.1
dns-server 8.8.8.8
exit
```

9. Konfigurasi IP static pada VPCS untuk VLAN 30 dan 40.

(PC VLAN 30) ip 192.168.30.10/24 192.168.30.1

(PC VLAN 40) ip 192.168.40.10/24 192.168.40.1

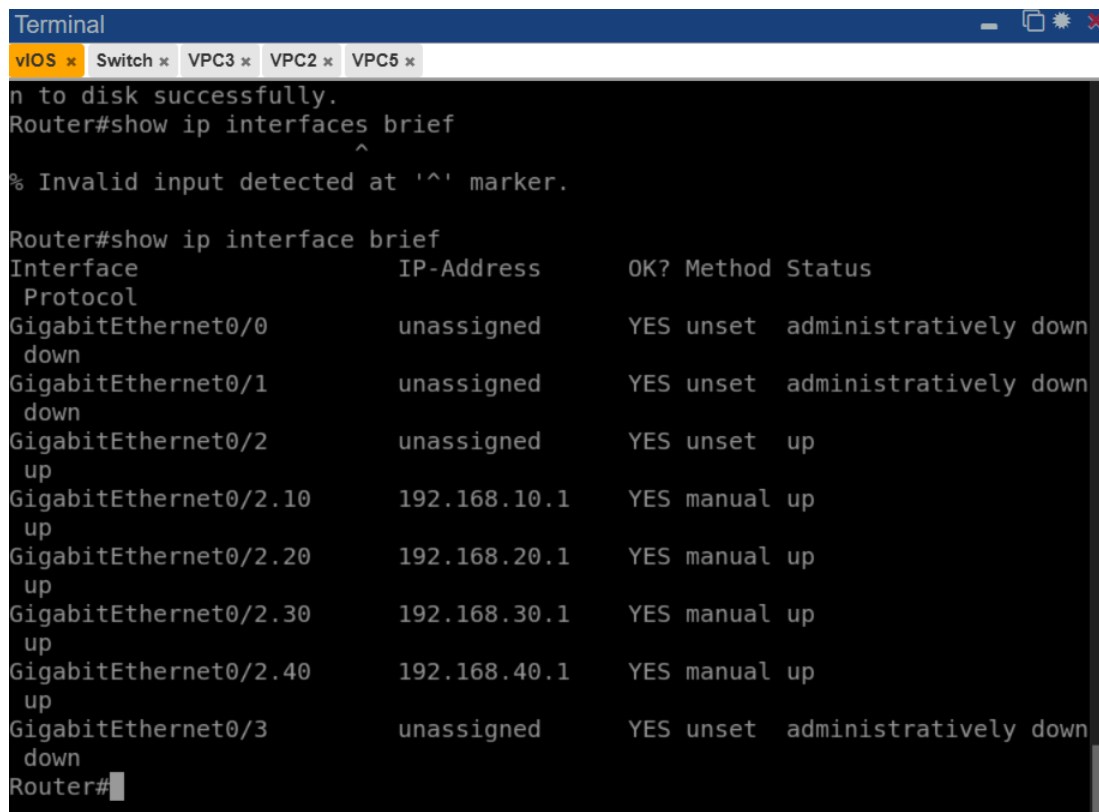
(PC VLAN 10 dan 20) ip dhcp

10. Verifikasi Cisco Router.

show ip interface brief

show ip dhcp pool

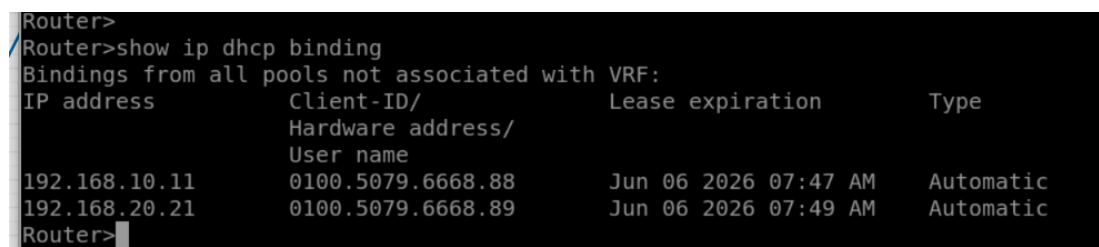
show ip dhcp binding



```
Terminal
vIOS x Switch x VPC3 x VPC2 x VPC5 x
n to disk successfully.
Router#show ip interfaces brief
^
% Invalid input detected at '^' marker.

Router#show ip interface brief
Interface                IP-Address      OK? Method Status
Protocol
GigabitEthernet0/0       unassigned      YES unset  administratively down
down
GigabitEthernet0/1       unassigned      YES unset  administratively down
down
GigabitEthernet0/2       unassigned      YES unset  up
up
GigabitEthernet0/2.10    192.168.10.1    YES manual  up
up
GigabitEthernet0/2.20    192.168.20.1    YES manual  up
up
GigabitEthernet0/2.30    192.168.30.1    YES manual  up
up
GigabitEthernet0/2.40    192.168.40.1    YES manual  up
up
GigabitEthernet0/3       unassigned      YES unset  administratively down
down
Router#
```

Gambar 3: Verifikasi interface Cisco Router dengan show ip interface brief



```
Router>
Router>show ip dhcp binding
Bindings from all pools not associated with VRF:
IP address      Client-ID/      Lease expiration      Type
Hardware address/
User name
192.168.10.11    0100.5079.6668.88 Jun 06 2026 07:47 AM  Automatic
192.168.20.21    0100.5079.6668.89 Jun 06 2026 07:49 AM  Automatic
Router>
```

Gambar 4: Verifikasi DHCP pool dan DHCP binding pada Cisco Router

11. Pengujian dari VPCS.

(PC VLAN 10) ip dhcp

(PC VLAN 10) show ip

```
(PC VLAN 10) ping 192.168.10.1
```

```
(PC VLAN 20) ip dhcp
```

```
(PC VLAN 20) show ip
```

```
(PC VLAN 20) ping 192.168.20.1
```

```
(PC VLAN 30) ping 192.168.30.1
```

```
(PC VLAN 40) ping 192.168.40.1
```

```
(PC VLAN 10) ping 192.168.20.11
```

```
(PC VLAN 10) ping 192.168.30.10
```

```
(PC VLAN 10) ping 192.168.40.10
```

```
show ip dhcp binding
```

1.3 Praktikum 3: Konfigurasi Huawei Router dan IP Address Antarrouter

Tujuan pada praktikum ini adalah mengonfigurasi LAN pada sisi Huawei, IP address link antarrouter, serta memverifikasi konektivitas antarrouter sebelum konfigurasi OSPF.

1. Konfigurasi LAN Huawei.

```
system-view
```

```
interface Ethernet1/0/2
```

```
description LAN-HUAWEI
```

```
ip address 192.168.50.1 255.255.255.0
```

```
undo shutdown
```

```
quit
```

```
commit
```

2. Konfigurasi IP VPC LAN Huawei dan verifikasi.

```
ip 192.168.50.10/24 192.168.50.1
```

```
show ip
```

```
ping 192.168.50.1
```



```

VPCS> ip 192.168.50.10/24 192.168.50.1
Checking for duplicate address...
PC1 : 192.168.50.10 255.255.255.0 gateway 192.168.50.1

VPCS> show ip

NAME       : VPCS[1]
IP/MASK     : 192.168.50.10/24
GATEWAY     : 192.168.50.1
DNS         :
MAC         : 00:50:79:66:68:87
LPORT      : 20000
RHOST:PORT  : 127.0.0.1:30000
MTU         : 1500

VPCS> █

```

Gambar 5: Verifikasi IP VPC LAN Huawei dengan show ip

```

VPCS> ping 192.168.50.1

84 bytes from 192.168.50.1 icmp_seq=1 ttl=255 time=7.322 ms
84 bytes from 192.168.50.1 icmp_seq=2 ttl=255 time=1.869 ms
84 bytes from 192.168.50.1 icmp_seq=3 ttl=255 time=1.570 ms
84 bytes from 192.168.50.1 icmp_seq=4 ttl=255 time=4.487 ms
84 bytes from 192.168.50.1 icmp_seq=5 ttl=255 time=2.819 ms

VPCS> █

```

Gambar 6: Pengujian ping dari VPC LAN Huawei ke gateway 192.168.50.1

3. Konfigurasi IP link Cisco – Huawei Link 1 (network 10.10.10.0/30).

(Cisco Router)

```

interface gi0/3
description LINK1-T0-HUAWEI
ip address 10.10.10.1 255.255.255.252
no shutdown
exit

```

(Huawei Router)

```

interface Ethernet1/0/0
description LINK1-T0-CISCO
ip address 10.10.10.2 255.255.255.252
undo shutdown
quit
commit

```

4. Konfigurasi IP link Cisco – Huawei Link 2 (network 10.10.11.0/30).

```
(Cisco Router)
interface gi0/0
description LINK2-T0-HUAWEI
ip address 10.10.11.1 255.255.255.252
no shutdown
exit
```

```
(Huawei Router)
interface Ethernet1/0/4
description LINK2-T0-CISCO
ip address 10.10.11.2 255.255.255.252
undo shutdown
quit
commit
```

5. Konfigurasi IP link Huawei – MikroTik (network 10.10.20.0/30).

```
(Huawei Router)
interface Ethernet1/0/3
description LINK-T0-MIKROTIK
ip address 10.10.20.1 255.255.255.252
undo shutdown
quit
commit
```

```
(MikroTik)
/ip address add address=10.10.20.2/30 interface=ether1 comment=T0-HUAWEI
```

6. Konfigurasi IP link Cisco – MikroTik (network 10.10.30.0/30).

```
(Cisco Router)
interface gi0/1
description LINK-T0-MIKROTIK
ip address 10.10.30.1 255.255.255.252
no shutdown
exit
```

```
(MikroTik)
/ip address add address=10.10.30.2/30 interface=ether2 comment=T0-CISCO
```

7. Verifikasi interface pada ketiga router.

```
(Cisco Router)    show ip interface brief
(Huawei Router)    display ip interface brief
(MikroTik)        /ip address print
```

```

Jun  5 10:32:38.048: %SYS-5-CONFIG-I-1: Configured from console by console
% Ambiguous command: "show i"
Router#show ip interface brief
Interface                                IP-Address      OK? Method Status
Protocol
GigabitEthernet0/0                      10.10.11.1      YES manual up
up
GigabitEthernet0/1                      10.10.30.1      YES manual up
up
GigabitEthernet0/2                      unassigned      YES NVRAM  up
up
GigabitEthernet0/2.10                   192.168.10.1    YES NVRAM  up
up
GigabitEthernet0/2.20                   192.168.20.1    YES NVRAM  up
up
GigabitEthernet0/2.30                   192.168.30.1    YES NVRAM  up
up
GigabitEthernet0/2.40                   192.168.40.1    YES NVRAM  up
up
GigabitEthernet0/3                      10.10.10.1      YES manual up
up
Router#

```

Gambar 7: Verifikasi interface Cisco Router

```

NE40E * VPC1 * Mikrotik * vIOS *

[~HUAWEI]display ip interface brief
*down: administratively down
!down: FIB overload down
^down: standby
(l): loopback
(s): spoofing
(d): Dampening Suppressed
(E): E-Trunk down
The number of interface that is UP in Physical is 23
The number of interface that is DOWN in Physical is 0
The number of interface that is UP in Protocol is 16
The number of interface that is DOWN in Protocol is 7

Interface                                IP Address/Mask    Physical
  Protocol VPN
Ethernet1/0/0                            10.10.10.2/30       up
  up    --
Ethernet1/0/0.4094                        128.1.138.137/16    up
  up    l3vpn
Ethernet1/0/1                            unassigned          up
  down  --
Ethernet1/0/1.4094                        128.1.138.137/16    up
  up    l3vpn
Ethernet1/0/2                            192.168.50.1/24     up
  up    --
Ethernet1/0/2.4094                        128.1.138.137/16    up
  up    l3vpn
Ethernet1/0/3                            10.10.20.1/30       up
  up    --
Ethernet1/0/3.4094                        128.1.138.137/16    up
  up    l3vpn
Ethernet1/0/4                            10.10.11.2/30       up
  up    --
Ethernet1/0/4.4094                        128.1.138.137/16    up
  up    l3vpn
Ethernet1/0/5                            unassigned          up
  down  --
Ethernet1/0/5.4094                        128.1.138.137/16    up
  up    l3vpn
Ethernet1/0/6                            unassigned          up
  down  --
Ethernet1/0/6.4094                        128.1.138.137/16    up
  up    l3vpn

```

Gambar 8: Verifikasi interface Huawei Router

```
comment=TO-CISCO
[admin@MikroTik] > /ip address print
Flags: X - disabled, I - invalid, D - dynamic
#   ADDRESS          NETWORK    INTERFACE
0   ;;; TO-HUAWEI
    10.10.20.2/30     10.10.20.0 ether1
1   ;;; TO-CISCO
    10.10.30.2/30     10.10.30.0 ether2
[admin@MikroTik] > █
```

Gambar 9: Verifikasi interface MikroTik

8. Verifikasi ping antarrouter.

(Cisco Router)

ping 10.10.10.2

ping 10.10.11.2

ping 10.10.30.2

(Huawei Router)

ping 10.10.10.1

ping 10.10.11.1

ping 10.10.20.2

(MikroTik)

ping 10.10.20.1

ping 10.10.30.1

```
Router#ping 10.10.10.2
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.10.10.2, timeout is 2 seconds:
.!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 1/1/3 ms
Router#ping 10.10.11.2
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.10.11.2, timeout is 2 seconds:
.!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 1/1/2 ms
Router#ping 10.10.30.2
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.10.30.2, timeout is 2 seconds:
.!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 1/1/2 ms
Router#█
```

Gambar 10: Verifikasi ping antarrouter dari Cisco Router

```

[~HUAWEI]ping 10.10.10.1
PING 10.10.10.1: 56 data bytes, press CTRL_C to break
  Reply from 10.10.10.1: bytes=56 Sequence=1 ttl=255 time=2 ms
  Reply from 10.10.10.1: bytes=56 Sequence=2 ttl=255 time=2 ms
  Reply from 10.10.10.1: bytes=56 Sequence=3 ttl=255 time=1 ms
  Reply from 10.10.10.1: bytes=56 Sequence=4 ttl=255 time=1 ms
  Reply from 10.10.10.1: bytes=56 Sequence=5 ttl=255 time=2 ms

--- 10.10.10.1 ping statistics ---
  5 packet(s) transmitted
  5 packet(s) received
  0.00% packet loss
  round-trip min/avg/max = 1/1/2 ms

[~HUAWEI]ping 10.10.20.2
PING 10.10.20.2: 56 data bytes, press CTRL_C to break
  Reply from 10.10.20.2: bytes=56 Sequence=1 ttl=64 time=1 ms
  Reply from 10.10.20.2: bytes=56 Sequence=2 ttl=64 time=1 ms
  Reply from 10.10.20.2: bytes=56 Sequence=3 ttl=64 time=1 ms
  Reply from 10.10.20.2: bytes=56 Sequence=4 ttl=64 time=1 ms
  Reply from 10.10.20.2: bytes=56 Sequence=5 ttl=64 time=1 ms

--- 10.10.20.2 ping statistics ---
  5 packet(s) transmitted
  5 packet(s) received
  0.00% packet loss
  round-trip min/avg/max = 1/1/1 ms

[~HUAWEI]ping 10.10.11.1
PING 10.10.11.1: 56 data bytes, press CTRL_C to break
  Reply from 10.10.11.1: bytes=56 Sequence=1 ttl=255 time=1 ms
  Reply from 10.10.11.1: bytes=56 Sequence=2 ttl=255 time=1 ms
  Reply from 10.10.11.1: bytes=56 Sequence=3 ttl=255 time=1 ms
  Reply from 10.10.11.1: bytes=56 Sequence=4 ttl=255 time=2 ms
  Reply from 10.10.11.1: bytes=56 Sequence=5 ttl=255 time=2 ms

--- 10.10.11.1 ping statistics ---
  5 packet(s) transmitted
  5 packet(s) received
  0.00% packet loss
  round-trip min/avg/max = 1/1/2 ms

[~HUAWEI]

```

Gambar 11: Verifikasi ping antarrouter dari Huawei Router

```
[admin@MikroTik] > ping 10.10.20.1
SEQ HOST                                SIZE TTL TIME  STATUS
0 10.10.20.1                            56 255 2ms
1 10.10.20.1                            56 255 1ms
2 10.10.20.1                            56 255 1ms
3 10.10.20.1                            56 255 0ms
4 10.10.20.1                            56 255 0ms
5 10.10.20.1                            56 255 0ms
6 10.10.20.1                            56 255 0ms
7 10.10.20.1                            56 255 1ms
8 10.10.20.1                            56 255 0ms
sent=9 received=9 packet-loss=0% min-rtt=0ms avg-rtt=0ms max-rtt=2ms

[admin@MikroTik] > ping 10.10.30.1
SEQ HOST                                SIZE TTL TIME  STATUS
0 10.10.30.1                            56 255 1ms
1 10.10.30.1                            56 255 0ms
2 10.10.30.1                            56 255 2ms
3 10.10.30.1                            56 255 0ms
4 10.10.30.1                            56 255 1ms
sent=5 received=5 packet-loss=0% min-rtt=0ms avg-rtt=0ms max-rtt=2ms

[admin@MikroTik] >
```

Gambar 12: Verifikasi ping antarrouter dari MikroTik

1.4 Praktikum 4: Konfigurasi OSPF Multivendor dan OSPF Cost

Tujuan pada praktikum ini adalah mengonfigurasi routing dinamis OSPF pada tiga perangkat dari vendor berbeda yaitu Cisco Router, Huawei Router, dan MikroTik, dengan dua link Cisco-Huawei sebagai jalur utama dan MikroTik sebagai jalur backup.

1. Konfigurasi OSPF pada Cisco Router.

```
enable
configure terminal

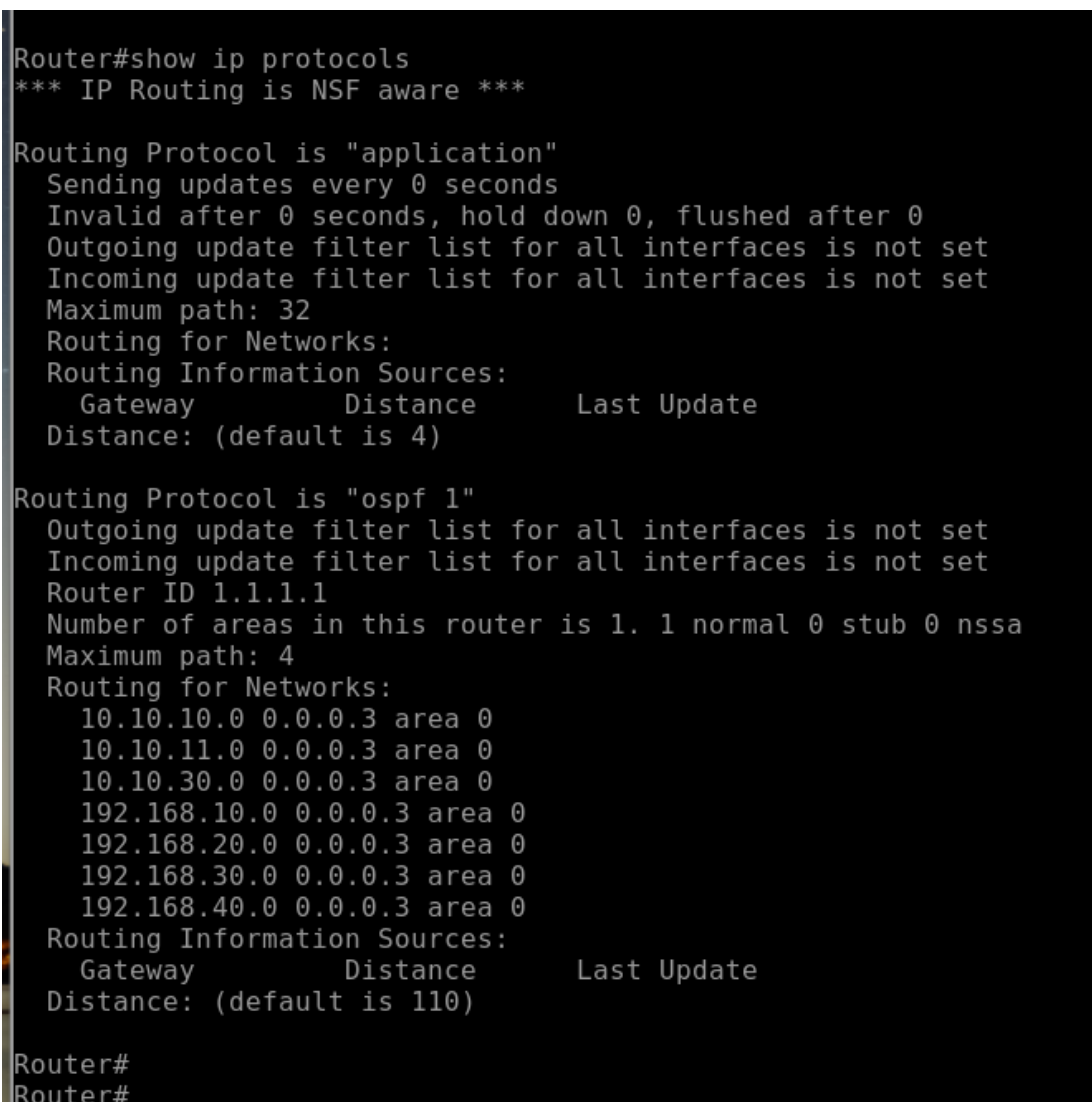
router ospf 1
router-id 1.1.1.1
network 10.10.10.0 0.0.0.3 area 0
network 10.10.11.0 0.0.0.3 area 0
network 10.10.30.0 0.0.0.3 area 0
network 192.168.10.0 0.0.0.255 area 0
network 192.168.20.0 0.0.0.255 area 0
network 192.168.30.0 0.0.0.255 area 0
network 192.168.40.0 0.0.0.255 area 0
exit

interface gi0/3
ip ospf cost 10
exit
interface gi0/0
ip ospf cost 10
```

```
exit
interface gi0/1
 ip ospf cost 100
exit
```

Verifikasi OSPF Cisco:

```
show ip ospf neighbor
show ip route ospf
show ip protocols
```



```
Router#show ip protocols
*** IP Routing is NSF aware ***

Routing Protocol is "application"
  Sending updates every 0 seconds
  Invalid after 0 seconds, hold down 0, flushed after 0
  Outgoing update filter list for all interfaces is not set
  Incoming update filter list for all interfaces is not set
  Maximum path: 32
  Routing for Networks:
  Routing Information Sources:
    Gateway          Distance          Last Update
  Distance: (default is 4)

Routing Protocol is "ospf 1"
  Outgoing update filter list for all interfaces is not set
  Incoming update filter list for all interfaces is not set
  Router ID 1.1.1.1
  Number of areas in this router is 1. 1 normal 0 stub 0 nssa
  Maximum path: 4
  Routing for Networks:
    10.10.10.0 0.0.0.3 area 0
    10.10.11.0 0.0.0.3 area 0
    10.10.30.0 0.0.0.3 area 0
    192.168.10.0 0.0.0.3 area 0
    192.168.20.0 0.0.0.3 area 0
    192.168.30.0 0.0.0.3 area 0
    192.168.40.0 0.0.0.3 area 0
  Routing Information Sources:
    Gateway          Distance          Last Update
  Distance: (default is 110)

Router#
Router#
```

Gambar 13: Verifikasi IP protocol pada Cisco Router

2. Konfigurasi OSPF pada Huawei Router.

```
system-view

ospf 1 router-id 2.2.2.2
 area 0.0.0.0
```



```

network 10.10.10.0 0.0.0.3
network 10.10.11.0 0.0.0.3
network 10.10.20.0 0.0.0.3
network 192.168.50.0 0.0.0.255
quit
quit

```

```

interface Ethernet1/0/0
  ospf cost 10
quit
interface Ethernet1/0/4
  ospf cost 10
quit
interface Ethernet1/0/3
  ospf cost 100
quit

```

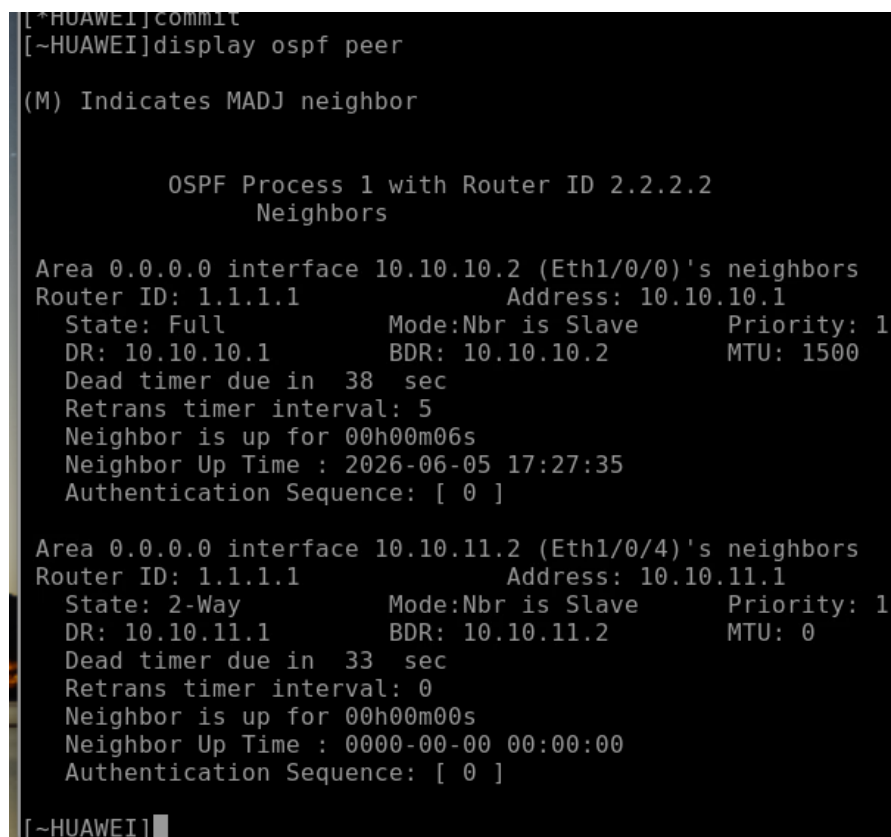
```
commit
```

Verifikasi OSPF Huawei:

```

display ospf peer
display ip routing-table protocol ospf
display ospf routing

```



```

[*HUAWEI]commit
[~HUAWEI]display ospf peer

(M) Indicates MADJ neighbor

          OSPF Process 1 with Router ID 2.2.2.2
          Neighbors

Area 0.0.0.0 interface 10.10.10.2 (Eth1/0/0)'s neighbors
Router ID: 1.1.1.1           Address: 10.10.10.1
  State: Full                Mode:Nbr is Slave      Priority: 1
  DR: 10.10.10.1            BDR: 10.10.10.2          MTU: 1500
  Dead timer due in 38 sec
  Retrans timer interval: 5
  Neighbor is up for 00h00m06s
  Neighbor Up Time : 2026-06-05 17:27:35
  Authentication Sequence: [ 0 ]

Area 0.0.0.0 interface 10.10.11.2 (Eth1/0/4)'s neighbors
Router ID: 1.1.1.1           Address: 10.10.11.1
  State: 2-Way               Mode:Nbr is Slave      Priority: 1
  DR: 10.10.11.1            BDR: 10.10.11.2          MTU: 0
  Dead timer due in 33 sec
  Retrans timer interval: 0
  Neighbor is up for 00h00m00s
  Neighbor Up Time : 0000-00-00 00:00:00
  Authentication Sequence: [ 0 ]

[~HUAWEI]

```

Gambar 14: Verifikasi OSPF peer pada Huawei Router

```

NE40E * vIOS * Switch * VPC1 * Mikrotik *
_public_ Routing Table : OSPF
Destinations : 9          Routes : 14

OSPF routing table status : <Active>
Destinations : 5          Routes : 10

Destination/Mask    Proto  Pre  Cost           Flags NextHop          Interfa
ce
10.10.30.0/30      OSPF   10   110            D    10.10.11.1        Etherne
t1/0/4
10.10.10.0/30      OSPF   10   110            D    10.10.10.1        Etherne
t1/0/0
192.168.10.0/24    OSPF   10   11             D    10.10.11.1        Etherne
t1/0/4
192.168.20.0/24    OSPF   10   11             D    10.10.10.1        Etherne
t1/0/0
192.168.30.0/24    OSPF   10   11             D    10.10.11.1        Etherne
t1/0/4
192.168.40.0/24    OSPF   10   11             D    10.10.10.1        Etherne
t1/0/0
192.168.50.0/24    OSPF   10   1             D    192.168.50.1      Etherne
t1/0/2

OSPF routing table status : <Inactive>
Destinations : 4          Routes : 4

Destination/Mask    Proto  Pre  Cost           Flags NextHop          Interfa
ce
10.10.10.0/30      OSPF   10   10             10.10.10.2        Etherne
t1/0/0
10.10.11.0/30      OSPF   10   10             10.10.11.2        Etherne
t1/0/4
10.10.20.0/30      OSPF   10   100            10.10.20.1        Etherne
t1/0/3
192.168.50.0/24    OSPF   10   1             192.168.50.1      Etherne
t1/0/2
[~HUAWEI]

```

Gambar 15: Verifikasi routing table OSPF pada Huawei Router

```
[~HUAWEI]display ospf routing

      OSPF Process 1 with Router ID 2.2.2.2
      Routing Tables

Routing for Network
Destination      Cost      Type      NextHop      AdvRouter      Area
10.10.10.0/30    10        Direct   10.10.10.2    2.2.2.2        0.0.0.0
10.10.11.0/30    10        Direct   10.10.11.2    2.2.2.2        0.0.0.0
10.10.20.0/30    100       Direct   10.10.20.1    2.2.2.2        0.0.0.0
10.10.30.0/30    110       Stub     10.10.11.1    1.1.1.1        0.0.0.0
10.10.30.0/30    110       Stub     10.10.10.1    1.1.1.1        0.0.0.0
192.168.10.0/24  11        Stub     10.10.11.1    1.1.1.1        0.0.0.0
192.168.10.0/24  11        Stub     10.10.10.1    1.1.1.1        0.0.0.0
192.168.20.0/24  11        Stub     10.10.11.1    1.1.1.1        0.0.0.0
192.168.20.0/24  11        Stub     10.10.10.1    1.1.1.1        0.0.0.0
192.168.30.0/24  11        Stub     10.10.11.1    1.1.1.1        0.0.0.0
192.168.30.0/24  11        Stub     10.10.10.1    1.1.1.1        0.0.0.0
192.168.40.0/24  11        Stub     10.10.11.1    1.1.1.1        0.0.0.0
192.168.40.0/24  11        Stub     10.10.10.1    1.1.1.1        0.0.0.0
192.168.50.0/24  1         Direct   192.168.50.1  2.2.2.2        0.0.0.0

Total Nets: 9
Intra Area: 9 Inter Area: 0 ASE: 0 NSSA: 0
---- More ----
```

Gambar 16: Verifikasi OSPF routing pada Huawei Router

3. Konfigurasi OSPF pada Mikrotik.

```
/routing ospf instance set default router-id=3.3.3.3
/routing ospf network add network=10.10.20.0/30 area=backbone
/routing ospf network add network=10.10.30.0/30 area=backbone
/routing ospf interface add interface=ether1 cost=100
/routing ospf interface add interface=ether2 cost=100
```

Verifikasi Mikrotik:

```
/routing ospf neighbor print
/ip route print where ospf
```

```
[admin@MikroTik] > /routing ospf neighbor print
0 instance=default router-id=1.1.1.1 address=10.10.30.1 interface=ether2 priority=1
  dr-address=10.10.30.1 backup-dr-address=10.10.30.2 state="Full" state-changes=5
  ls-retransmits=0 ls-requests=0 db-summaries=0 adjacency=16s

1 instance=default router-id=2.2.2.2 address=10.10.20.1 interface=ether1 priority=1
  dr-address=10.10.20.1 backup-dr-address=10.10.20.2 state="Full" state-changes=5
  ls-retransmits=0 ls-requests=0 db-summaries=0 adjacency=25s
[admin@MikroTik] > █
```

Gambar 17: Verifikasi OSPF neighbor pada Mikrotik

```
cs-retransmits=0 cs-requests=0 db-summaries=0 adjacency=255
[admin@MikroTik] > /ip route print where ospf
Flags: X - disabled, A - active, D - dynamic,
C - connect, S - static, r - rip, b - bgp, o - ospf, m - mme,
B - blackhole, U - unreachable, P - prohibit
# DST-ADDRESS PREF-SRC GATEWAY DISTANCE
0 ADo 10.10.10.0/30 10.10.20.1 110
10.10.30.1
1 ADo 10.10.11.0/30 10.10.20.1 110
10.10.30.1
2 ADo 192.168.10.0/24 10.10.30.1 110
3 ADo 192.168.20.0/24 10.10.30.1 110
4 ADo 192.168.30.0/24 10.10.30.1 110
5 ADo 192.168.40.0/24 10.10.30.1 110
6 ADo 192.168.50.0/24 10.10.20.1 110
[admin@MikroTik] >
```

Gambar 18: Verifikasi routing table OSPF pada MikroTik

4. Verifikasi OSPF neighbor pada ketiga router.

(Cisco Router) show ip ospf neighbor

(Huawei Router) display ospf peer

(MikroTik) /routing ospf neighbor print

```
*****
Router>show ip ospf neighbor
Neighbor ID Pri State Dead Time Address Interface
3.3.3.3 1 FULL/BDR 00:00:33 10.10.30.2 GigabitEthernet0/1
2.2.2.2 1 FULL/BDR 00:00:30 10.10.11.2 GigabitEthernet0/0
2.2.2.2 1 FULL/BDR 00:00:34 10.10.10.2 GigabitEthernet0/3
Router>
```

Gambar 19: OSPF neighbor pada Cisco Router

```
[~HUAWEI]
[~HUAWEI]display ospf peer

(M) Indicates MADJ neighbor

                OSPF Process 1 with Router ID 2.2.2.2
                  Neighbors

Area 0.0.0.0 interface 10.10.10.2 (Eth1/0/0)'s neighbors
Router ID: 1.1.1.1           Address: 10.10.10.1
  State: Full                Mode:Nbr is Slave      Priority: 1
  DR: 10.10.10.1            BDR: 10.10.10.2         MTU: 1500
  Dead timer due in 39 sec
  Retrans timer interval: 5
  Neighbor is up for 00h06m08s
  Neighbor Up Time : 2026-06-05 17:27:35
  Authentication Sequence: [ 0 ]

Area 0.0.0.0 interface 10.10.20.1 (Eth1/0/3)'s neighbors
Router ID: 3.3.3.3           Address: 10.10.20.2
  State: Full                Mode:Nbr is Master    Priority: 1
  DR: 10.10.20.1            BDR: 10.10.20.2         MTU: 1500
  Dead timer due in 37 sec
  Retrans timer interval: 5
  Neighbor is up for 00h01m48s
  Neighbor Up Time : 2026-06-05 17:31:56
  Authentication Sequence: [ 0 ]

Area 0.0.0.0 interface 10.10.11.2 (Eth1/0/4)'s neighbors
Router ID: 1.1.1.1           Address: 10.10.11.1
  State: Full                Mode:Nbr is Slave      Priority: 1
  DR: 10.10.11.1            BDR: 10.10.11.2         MTU: 1500
  Dead timer due in 35 sec
  Retrans timer interval: 5
  Neighbor is up for 00h05m32s
  Neighbor Up Time : 2026-06-05 17:28:13
  Authentication Sequence: [ 0 ]

[~HUAWEI]
```

Gambar 20: OSPF neighbor pada Huawei Router

```
[admin@MikroTik] > /routing ospf neighbor print
0 instance=default router-id=1.1.1.1 address=10.10.30.1 interface=ether2 priority=1
  dr-address=10.10.30.1 backup-dr-address=10.10.30.2 state="Full" state-changes=5
  ls-retransmits=0 ls-requests=0 db-summaries=0 adjacency=2m22s

1 instance=default router-id=2.2.2.2 address=10.10.20.1 interface=ether1 priority=1
  dr-address=10.10.20.1 backup-dr-address=10.10.20.2 state="Full" state-changes=5
  ls-retransmits=0 ls-requests=0 db-summaries=0 adjacency=2m31s
[admin@MikroTik] >
```

Gambar 21: OSPF neighbor pada MikroTik

5. Verifikasi routing table pada ketiga router.

(Cisco Router)

```
show ip route 192.168.50.0
```

```
show ip route ospf
```

(Huawei Router)

```
display ip routing-table protocol ospf
```

(MikroTik)

```
/ip route print where ospf
```

```
Router#show ip route 192.168.50.0
Routing entry for 192.168.50.0/24
  Known via "ospf 1", distance 110, metric 11, type intra area
  Last update from 10.10.10.2 on GigabitEthernet0/3, 00:43:08 ago
  Routing Descriptor Blocks:
    * 10.10.11.2, from 2.2.2.2, 00:43:18 ago, via GigabitEthernet0/0
      Route metric is 11, traffic share count is 1
    10.10.10.2, from 2.2.2.2, 00:43:08 ago, via GigabitEthernet0/3
      Route metric is 11, traffic share count is 1
```

Gambar 22: Routing table Cisco Router menuju LAN Huawei (show ip route 192.168.50.0)

```
Router#show ip route ospf
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PFR

Gateway of last resort is not set

    10.0.0.0/8 is variably subnetted, 7 subnets, 2 masks
O       10.10.20.0/30 [110/110] via 10.10.11.2, 00:43:20, GigabitEthernet0/0
        [110/110] via 10.10.10.2, 00:43:10, GigabitEthernet0/3
O       192.168.50.0/24 [110/11] via 10.10.11.2, 00:43:20, GigabitEthernet0/0
        [110/11] via 10.10.10.2, 00:43:10, GigabitEthernet0/3
```

Gambar 23: Semua route OSPF pada Cisco Router (show ip route ospf)

```
[~HUAWEI]display ip routing-table protocol ospf
Route Flags: R - relay, D - download to fib, T - to vpn-instance, B - black hole route
-----
_public_ Routing Table : OSPF
Destinations : 9          Routes : 14

OSPF routing table status : <Active>
Destinations : 5          Routes : 10

Destination/Mask    Proto   Pre  Cost           Flags NextHop         Interface
-----
10.10.30.0/30       OSPF    10   110            D  10.10.11.1        Ethernet1/0/4
10.10.10.0/30       OSPF    10   110            D  10.10.10.1        Ethernet1/0/0
192.168.10.0/24     OSPF    10   11             D  10.10.11.1        Ethernet1/0/4
10.10.10.0/30       OSPF    10   11             D  10.10.10.1        Ethernet1/0/0
192.168.20.0/24     OSPF    10   11             D  10.10.11.1        Ethernet1/0/4
10.10.10.0/30       OSPF    10   11             D  10.10.10.1        Ethernet1/0/0
192.168.30.0/24     OSPF    10   11             D  10.10.11.1        Ethernet1/0/4
10.10.10.0/30       OSPF    10   11             D  10.10.10.1        Ethernet1/0/0
192.168.40.0/24     OSPF    10   11             D  10.10.11.1        Ethernet1/0/4
10.10.10.0/30       OSPF    10   11             D  10.10.10.1        Ethernet1/0/0

OSPF routing table status : <Inactive>
Destinations : 4          Routes : 4

---- More ----

          OSPF    10   11             D  10.10.10.1        Ethernet1/0/0

OSPF routing table status : <Inactive>
Destinations : 4          Routes : 4

Destination/Mask    Proto   Pre  Cost           Flags NextHop         Interface
-----
10.10.10.0/30       OSPF    10   10             10.10.10.2        Ethernet1/0/0
10.10.11.0/30       OSPF    10   10             10.10.11.2        Ethernet1/0/4
10.10.20.0/30       OSPF    10  100            10.10.20.1        Ethernet1/0/3
192.168.50.0/24     OSPF    10   1              192.168.50.1      Ethernet1/0/2
```

Gambar 24: Routing table OSPF pada Huawei Router

```
[admin@MikroTik] > /ip route print where ospf
Flags: X - disabled, A - active, D - dynamic, C - connect, S - static, r - rip, b - bgp, o - ospf, m - mme,
B - blackhole, U - unreachable, P - prohibit
# DST-ADDRESS      PREF-SRC  GATEWAY          DISTANCE
0 ADo 10.10.10.0/30          10.10.30.1       110
10.10.20.1
1 ADo 10.10.11.0/30         10.10.30.1       110
10.10.20.1
2 ADo 192.168.10.0/24         10.10.30.1       110
3 ADo 192.168.20.0/24         10.10.30.1       110
4 ADo 192.168.30.0/24         10.10.30.1       110
5 ADo 192.168.40.0/24         10.10.30.1       110
6 ADo 192.168.50.0/24         10.10.20.1       110
```

Gambar 25: Routing table OSPF pada MikroTik

6. Verifikasi koneksi end-to-end.

(PC VLAN 10) ping 192.168.50.10

(PC VLAN 20) ping 192.168.50.10

(PC VLAN 30) ping 192.168.50.10

(PC VLAN 40) ping 192.168.50.10

(PC LAN Huawei)

ping 192.168.10.11

ping 192.168.20.21

ping 192.168.30.10

ping 192.168.40.10

```

0;VPC2
VPCS>
VPCS> ping 192.168.50.10

84 bytes from 192.168.50.10 icmp_seq=1 ttl=62 time=6.210 ms
84 bytes from 192.168.50.10 icmp_seq=2 ttl=62 time=5.697 ms
84 bytes from 192.168.50.10 icmp_seq=3 ttl=62 time=4.306 ms
84 bytes from 192.168.50.10 icmp_seq=4 ttl=62 time=3.122 ms
84 bytes from 192.168.50.10 icmp_seq=5 ttl=62 time=2.886 ms

VPCS>

```

Gambar 26: Ping dari PC VLAN 10 ke LAN Huawei

```

0;VPC3
VPCS>
VPCS> ping 192.168.50.10

84 bytes from 192.168.50.10 icmp_seq=1 ttl=62 time=3.872 ms
84 bytes from 192.168.50.10 icmp_seq=2 ttl=62 time=6.328 ms
84 bytes from 192.168.50.10 icmp_seq=3 ttl=62 time=3.406 ms
84 bytes from 192.168.50.10 icmp_seq=4 ttl=62 time=4.220 ms
84 bytes from 192.168.50.10 icmp_seq=5 ttl=62 time=2.996 ms

VPCS>

```

Gambar 27: Ping dari PC VLAN 20 ke LAN Huawei

```

0;VPC4
VPCS> ping 192.168.50.10

84 bytes from 192.168.50.10 icmp_seq=1 ttl=62 time=5.240 ms
84 bytes from 192.168.50.10 icmp_seq=2 ttl=62 time=4.478 ms
84 bytes from 192.168.50.10 icmp_seq=3 ttl=62 time=3.238 ms
84 bytes from 192.168.50.10 icmp_seq=4 ttl=62 time=8.046 ms
84 bytes from 192.168.50.10 icmp_seq=5 ttl=62 time=3.633 ms

```

Gambar 28: Ping dari PC VLAN 30 ke LAN Huawei

```

0;VPC5
VPCS> ping 192.168.50.10

84 bytes from 192.168.50.10 icmp_seq=1 ttl=62 time=12.165 ms
84 bytes from 192.168.50.10 icmp_seq=2 ttl=62 time=3.819 ms
84 bytes from 192.168.50.10 icmp_seq=3 ttl=62 time=4.111 ms
84 bytes from 192.168.50.10 icmp_seq=4 ttl=62 time=6.187 ms
84 bytes from 192.168.50.10 icmp_seq=5 ttl=62 time=4.899 ms

```

Gambar 29: Ping dari PC VLAN 40 ke LAN Huawei


```

VPCS> ping 192.168.20.21

84 bytes from 192.168.20.21 icmp_seq=1 ttl=62 time=14.491 ms
84 bytes from 192.168.20.21 icmp_seq=2 ttl=62 time=4.771 ms
84 bytes from 192.168.20.21 icmp_seq=3 ttl=62 time=3.873 ms
84 bytes from 192.168.20.21 icmp_seq=4 ttl=62 time=6.457 ms
84 bytes from 192.168.20.21 icmp_seq=5 ttl=62 time=5.287 ms

VPCS> ping 192.168.30.10

84 bytes from 192.168.30.10 icmp_seq=1 ttl=62 time=7.776 ms
84 bytes from 192.168.30.10 icmp_seq=2 ttl=62 time=3.756 ms
84 bytes from 192.168.30.10 icmp_seq=3 ttl=62 time=4.429 ms
84 bytes from 192.168.30.10 icmp_seq=4 ttl=62 time=4.725 ms
84 bytes from 192.168.30.10 icmp_seq=5 ttl=62 time=3.184 ms

VPCS> ping 192.168.40.10

84 bytes from 192.168.40.10 icmp_seq=1 ttl=62 time=8.279 ms
84 bytes from 192.168.40.10 icmp_seq=2 ttl=62 time=3.557 ms
84 bytes from 192.168.40.10 icmp_seq=3 ttl=62 time=6.948 ms
84 bytes from 192.168.40.10 icmp_seq=4 ttl=62 time=3.733 ms
84 bytes from 192.168.40.10 icmp_seq=5 ttl=62 time=3.155 ms

VPCS> ping 192.168.10.11

84 bytes from 192.168.10.11 icmp_seq=1 ttl=62 time=4.269 ms
84 bytes from 192.168.10.11 icmp_seq=2 ttl=62 time=4.426 ms
84 bytes from 192.168.10.11 icmp_seq=3 ttl=62 time=5.481 ms
84 bytes from 192.168.10.11 icmp_seq=4 ttl=62 time=3.163 ms
84 bytes from 192.168.10.11 icmp_seq=5 ttl=62 time=3.510 ms

```

Gambar 30: Ping dari PC LAN Huawei ke semua VLAN

1.5 Praktikum 5: Pengujian Failover OSPF

Tujuan pada praktikum ini adalah memverifikasi bahwa OSPF dapat berpindah jalur secara otomatis ketika link utama mengalami gangguan.

1. Kondisi normal: dua link Cisco-Huawei aktif. Cek route dari Cisco menuju LAN Huawei dan lakukan trace.

(Cisco Router) `show ip route 192.168.50.0`

(PC VLAN 10) `trace 192.168.50.10`

```

Router#show ip route 192.168.50.0
Routing entry for 192.168.50.0/24
  Known via "ospf 1", distance 110, metric 11, type intra area
  Last update from 10.10.11.2 on GigabitEthernet0/0, 00:27:36 ago
  Routing Descriptor Blocks:
    10.10.11.2, from 2.2.2.2, 00:27:36 ago, via GigabitEthernet0/0
      Route metric is 11, traffic share count is 1
    * 10.10.10.2, from 2.2.2.2, 00:27:46 ago, via GigabitEthernet0/3
      Route metric is 11, traffic share count is 1
Router#

```

Gambar 31: Show IP Route normal

```

0;VPC2
VPCS> trace 192.168.50.10
trace to 192.168.50.10, 8 hops max, press Ctrl+C to stop
 1  192.168.10.1    3.845 ms  1.619 ms  2.376 ms
 2  10.10.10.2     9.289 ms  3.516 ms  3.537 ms
 3  *192.168.50.10 6.403 ms (ICMP type:3, code:3, Destination port unreachable)

```

Gambar 32: Trace kondisi normal — traffic melewati jalur langsung Cisco ke Huawei

2. Kondisi 1 link Cisco-Huawei mati. Matikan salah satu link, cek route, dan lakukan trace.

(Cisco Router)

```
configure terminal
```

```
interface gi0/3
```

```
shutdown
```

```
exit
```

```
show ip route 192.168.50.0
```

(PC VLAN 10)

```
trace 192.168.50.10
```

```

Router#show ip route 192.168.50.0
Routing entry for 192.168.50.0/24
  Known via "ospf 1", distance 110, metric 11, type intra area
  Last update from 10.10.11.2 on GigabitEthernet0/0, 00:31:20 ago
  Routing Descriptor Blocks:
    * 10.10.11.2, from 2.2.2.2, 00:31:20 ago, via GigabitEthernet0/0
      Route metric is 11, traffic share count is 1
Router#

```

Gambar 33: Show IP Route 1 link

```

VPCS> trace 192.168.50.10
trace to 192.168.50.10, 8 hops max, press Ctrl+C to stop
 1  192.168.10.1    1.897 ms  1.785 ms  3.314 ms
 2  10.10.11.2     5.529 ms  3.356 ms  2.511 ms
 3  *192.168.50.10 5.597 ms (ICMP type:3, code:3, Destination port unreachable)

```

Gambar 34: Trace saat 1 link mati — traffic tetap melewati jalur langsung Cisco ke Huawei

3. Kondisi dua link Cisco-Huawei mati. Matikan link kedua, cek route, dan lakukan trace.

(Cisco Router)

```
configure terminal
```

```
interface gi0/0
```

```
shutdown
```

```
exit
```

```
show ip route 192.168.50.0
```

(PC VLAN 10)

```
trace 192.168.50.10
```

```

Router#show ip route 192.168.50.0
Routing entry for 192.168.50.0/24
  Known via "ospf 1", distance 110, metric 11, type intra area
  Last update from 10.10.11.2 on GigabitEthernet0/0, 00:01:02 ago
  Routing Descriptor Blocks:
    10.10.11.2, from 2.2.2.2, 00:01:02 ago, via GigabitEthernet0/0
      Route metric is 11, traffic share count is 1
    * 10.10.10.2, from 2.2.2.2, 00:01:02 ago, via GigabitEthernet0/3
      Route metric is 11, traffic share count is 1
Router#

```

Gambar 35: Show IP Route 0 link

```

VPCS> trace 192.168.50.10
trace to 192.168.50.10, 8 hops max, press Ctrl+C to stop
 1  192.168.10.1    1.723 ms  1.402 ms  1.812 ms
 2  10.10.30.2     2.292 ms  1.630 ms  1.523 ms
 3  10.10.20.1     4.615 ms  2.606 ms  5.121 ms
 4  *192.168.50.10 4.975 ms (ICMP type:3, code:3, Destination port unreachable)

```

Gambar 36: Trace saat dua link mati — traffic berpindah ke jalur backup melalui MikroTik

4. Mengaktifkan kembali dua link Cisco-Huawei dan verifikasi jalur routing kembali ke link utama.

```

(Cisco Router)
configure terminal
interface gi0/3
  no shutdown
exit
interface gi0/0
  no shutdown
exit

show ip route 192.168.50.0

```

```

(PC VLAN 10)
trace 192.168.50.10

```

```

Router#show ip route 192.168.50.0
Routing entry for 192.168.50.0/24
  Known via "ospf 1", distance 110, metric 11, type intra area
  Last update from 10.10.11.2 on GigabitEthernet0/0, 00:27:36 ago
  Routing Descriptor Blocks:
    10.10.11.2, from 2.2.2.2, 00:27:36 ago, via GigabitEthernet0/0
      Route metric is 11, traffic share count is 1
    * 10.10.10.2, from 2.2.2.2, 00:27:46 ago, via GigabitEthernet0/3
      Route metric is 11, traffic share count is 1
Router#

```

Gambar 37: Show IP Route normal

```

0;VPC2
VPCS> trace 192.168.50.10
Trace to 192.168.50.10, 8 hops max, press Ctrl+C to stop
 1  192.168.10.1    3.845 ms  1.619 ms  2.376 ms
 2  10.10.10.2     9.289 ms  3.516 ms  3.537 ms
 3  *192.168.50.10 6.403 ms (ICMP type:3, code:3, Destination port unreachable)

```

Gambar 38: Trace kondisi normal — traffic melewati jalur langsung Cisco ke Huawei

2 Analisis Hasil Percobaan

Praktikum Modul 5 membahas implementasi VLAN, Trunk, OSPF Multivendor, dan pengujian failover pada topologi jaringan yang terdiri dari Cisco Switch, Cisco Router, Huawei Router, dan MikroTik. Praktikum dikerjakan secara berkelompok dengan menggunakan satu PC yang dioperasikan bergantian oleh anggota kelompok melalui server PNETLab.

Pengerjaan dimulai dari topologi yang telah disiapkan pada tugas pendahuluan sebelumnya. Pada awal praktikum yaitu Praktikum 1 dan 2, kelompok mengalami beberapa kendala. Terjadi ketidakonsistenan dalam penentuan VLAN 10, 20, 30, dan 40 pada konfigurasi access port sehingga hasil konfigurasi tidak sesuai dengan yang diharapkan. Akibatnya, kelompok memutuskan untuk melakukan wipe pada Cisco Switch dan mengatur ulang seluruh konfigurasi dari awal mengikuti petunjuk dari modul. Proses ini memakan waktu cukup lama sehingga kelompok terhambat di Praktikum 1 dan 2 dan kehabisan waktu saat praktikum berlangsung.

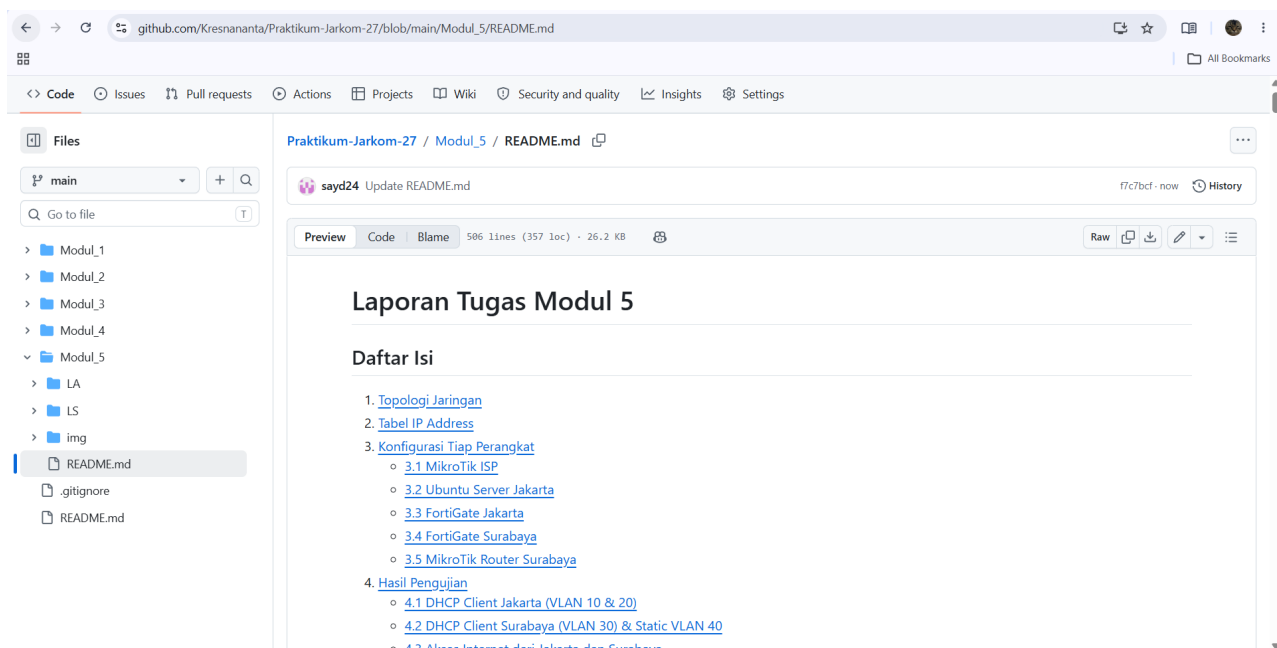
Kelompok terpaksa berhenti pada Praktikum 3 ketika baru akan merestart Huawei Router yang proses bootingnya membutuhkan waktu lama. Praktikum kemudian dilanjutkan di luar jam praktikum oleh salah satu anggota kelompok hingga menyelesaikan Praktikum 4 bagian 7 yaitu verifikasi routing table. Setelah itu, anggota lain melanjutkan pengerjaan dari bagian tersebut hingga seluruh praktikum selesai.

Hasil akhir praktikum menunjukkan bahwa seluruh konfigurasi berhasil berjalan dengan baik. VLAN 10, 20, 30, dan 40 berhasil dibuat dan diverifikasi pada Cisco Switch dengan trunk yang aktif membawa keempat VLAN. Cisco Router berhasil dikonfigurasi sebagai gateway dengan subinterfa-
ce ROAS dan DHCP server untuk VLAN 10 dan 20. OSPF berhasil terbentuk antara ketiga router dari vendor berbeda yaitu Cisco, Huawei, dan MikroTik dengan semua neighbor dalam kondisi Full. Pengujian end-to-end menunjukkan semua PC VLAN dapat saling berkomunikasi dengan PC LAN Huawei. Pengujian failover OSPF pada Praktikum 5 juga berhasil membuktikan bahwa ketika dua link utama Cisco-Huawei dimatikan, traffic secara otomatis berpindah ke jalur backup melalui MikroTik, dan kembali ke jalur utama ketika link diaktifkan kembali.

3 Hasil Tugas Modul

Tugas modul ini mensimulasikan arsitektur jaringan enterprise terdistribusi multi-site (HQ Jakarta dan Branch Surabaya) menggunakan platform PNETLab. Simulasi ini mengintegrasikan perangkat multi-vendor yang mencakup FortiGate, Cisco Router, MikroTik Router, Cisco Switch, serta Ubuntu Server dengan menerapkan berbagai teknologi seperti VRRP untuk redundansi gateway, ISC-DHCP Server terpusat, GRE Tunnel, dan OSPF dynamic routing over GRE. Seluruh dokumentasi sistem yang mencakup topologi jaringan, tabel pengalamatan IP, detail konfigurasi tiap perangkat, analisis performa,

hingga pembuktian hasil pengujian konektivitas end-to-end lintas-site telah dirangkum secara terpusat pada berkas *README* pada folder Modul 5 di repositori GitHub.



Gambar 39: Dokumentasi lengkap konfigurasi, analisis, dan pengujian Tugas Modul 5 pada berkas README repositori GitHub

4 Kesimpulan

Praktikum Modul 5 berhasil dilaksanakan dengan hasil yang sesuai dengan tujuan praktikum. VLAN berhasil diimplementasikan untuk memisahkan jaringan secara logis ke dalam empat segmen yaitu Finance, HR, IT, dan Marketing. Trunk port berhasil dikonfigurasi untuk membawa lalu lintas dari keempat VLAN melalui satu link fisik antara switch dan router. OSPF berhasil dikonfigurasi pada tiga perangkat dari vendor berbeda yaitu Cisco, Huawei, dan MikroTik sehingga semua router dapat saling bertukar informasi routing secara dinamis. Pengujian failover membuktikan bahwa OSPF mampu berpindah jalur secara otomatis dari link utama ke jalur backup melalui MikroTik ketika kedua link Cisco-Huawei dimatikan, dan kembali ke jalur utama ketika link dipulihkan. Secara keseluruhan, praktikum ini memberikan pemahaman yang baik tentang implementasi VLAN, ROAS, dan OSPF multivendor dalam jaringan nyata.

5 Lampiran

5.1 Dokumentasi saat Praktikum



Gambar 40: Dokumentasi pelaksanaan praktikum